ThmSmith Stage 0 derivation: `pid_mediation_proxy` / v1

Finite LP Bounds and Rank-One Proxy Failure for Natural Mediation Effects

ThmSmith Stage 0 solver

May 14, 2026

Abstract

This note studies the partial-identification problem for the natural-effect vector $\theta = (\mathrm{PNDE}, \mathrm{TNIE})$ in a binary mediation design with binary negative-control proxies. There is no regression estimator in this anchor; the estimand-defining object is a finite proxy-response linear program whose projection onto $(\mathrm{PNDE}, \mathrm{TNIE})$ gives the sharp identified set. The Stage -1 bridge formula conjectured a nontrivial rank-one affine zonotope generated by null directions of $K_{a,m,x}(z, w) = P(W = w \mid A = a, M = m, Z = z, X = x)$. The main conclusion is negative: for the stated observable bridge matrices, every rank-one bridge-null loading is zero, and the conjectured bridge-zonotope is not even an outer bound for the causal identified set without an additional latent bridge or completeness restriction. A finite response-type LP remains sharp, and the proposed proxy-separation idea is valid only after it is attached to that LP rather than to the refuted bridge-zonotope formula.

1 Introduction

Natural direct and indirect effects decompose a treatment effect into the part that bypasses a mediator and the part transmitted through it. This decomposition is often the scientific target, but it is fragile because the mediator-outcome relation can be confounded by variables that are not directly observed. Negative controls and proximal methods are designed for this situation: instead of assuming away the latent confounder, they use observed proxies to learn enough about it to identify or bound causal quantities.

The Stage -1 proposal for this instance asks whether the failure of proximal completeness has a simple sharp geometry. In the binary proxy setting it conjectures that rank-one bridge matrices generate a two-dimensional affine zonotope for the joint natural direct and indirect effects, with coordinate intervals obtained by projecting that zonotope. This would be useful if true, because weak or collinear proxies could still give closed-form sharp bounds rather than forcing a generic finite linear program.

The calculation below shows that the conjectured closed form does not hold for the causal identified set under the locked assumptions. There are two separate issues. First, since $K_{a,m,x}$ is a conditional probability matrix, the observed marginal $P(W \mid A = a, M = m, X = x)$ is a convex combination of its rows; therefore every bridge-null direction has zero loading in the proposed plug-in natural-effect functional. Second, the observable bridge equation $K_{a,m,x}h_{a,m,x} = q_{a,m,x}$ is not a latent bridge condition. Without a completeness or latent-bridge assumption, two finite causal laws can agree on the entire observed distribution and on all proxy restrictions while having different natural effects.

The useful replacement is the standard finite partial-identification object: a proxy-response linear program. It gives the sharp joint identified set by projecting the finite feasible response-type polytope onto the PNDE and TNIE coordinates. This note also records the part of the Stage -1 tightening idea that survives: strict improvement over no-proxy bounds is a finite convex separation statement for LP endpoint faces.

2 Related work and positioning

Robins and Greenland [Robins and Greenland(1992)] and Pearl [Pearl(2001)] introduced the counterfactual direct and indirect effects used here. Avin, Shpitser, and Pearl [Avin, Shpitser, and Pearl(2005)] showed that path-specific effects can fail to be identified when intermediate variables and hidden confounding interact, which is the reason this note treats the target as partially identified. Imai, Keele, and Yamamoto [Imai, Keele, and Yamamoto(2010)] give the familiar mediation formula under sequential ignorability; the present setting deliberately replaces that mediator-outcome ignorability condition by latent-confounding and proxy restrictions.

Dukes, Shpitser, and Tchetgen Tchetgen [Dukes, Shpitser, and Tchetgen Tchetgen(2023)] are the closest point-identification reference: proximal mediation analysis identifies natural effects using bridge functions when the required proxy bridge conditions are strong enough. Zhang, Li, Miao, and Tchetgen Tchetgen [Zhang, Li, Miao, and Tchetgen Tchetgen(2023)] show that bridge nonuniqueness need not always destroy identification of a target functional; this note explains why the particular rank-one formula proposed for the present mediation target is degenerate as a bridge-image calculation and insufficient as a causal bound. Ghassami, Shpitser, and Tchetgen Tchetgen [Ghassami, Shpitser, and Tchetgen Tchetge develop proxy-based partial-identification bounds without completeness; the corrected LP object here is in that same finite-proxy partial-identification spirit, but for the joint natural-effect pair.

The convex-analytic part follows the finite response-type tradition of Manski [Manski(1990)] and Balke and Pearl [Balke and Pearl(1997)]: impose all observable and structural restrictions as linear constraints, then project a polytope onto the causal functional of interest. Cai, Kuroki, Pearl, and Tian [Cai, Kuroki, Pearl, and Tian(2008)] apply symbolic LP methods to direct-effect bounds with a confounded intermediate variable. Miles, Kanki, Meloni, and Tchetgen Tchetgen [Miles, Kanki, Meloni, and Tchetgen Tchetgen(2017)] derive partial-identification bounds for the natural indirect effect, and Gabriel, Sachs, and Sjolander [Gabriel, Sachs, and Sjolander(2023)] emphasize that sharp bounds for decomposition components cannot generally be combined coordinatewise. Ding and VanderWeele [Ding and VanderWeele(2016)] provide sharp mediation sensitivity bounds; their work is a no-proxy comparator rather than a negative-control proxy result.

The local Stage -1 proposal is the immediate in-repository object positioned by this note. Its novelty claim was a rank-deficient proxy-null zonotope and an observable strict-tightening certificate. The first claim is refuted as a causal sharp set; the second is retained only as a corrected endpoint-face separation criterion for the finite LP sharp bound.

3 Formal setup

Let P be the observable law of (A, M, Y, Z, W, X), where $A, M, Y, Z, W \in \{0, 1\}$ and X has finite support $\mathcal{X}$. Write

$$p(a, m, y, z, w, x) = P(A = a, M = m, Y = y, Z = z, W = w, X = x).$$

The latent causal law is denoted $P^\star$. It contains a finite latent post-treatment mediator-outcome confounder U, potential mediators $M(a)$, and potential outcomes $Y(a, m)$. The factual variables obey the consistency relations stated below. All expectations are under $P^\star$ unless the argument explicitly says they are observable functionals of P.

The target parameter is the two-dimensional vector

$$\theta(P^\star) = \big(\mathrm{PNDE}(P^\star), \mathrm{TNIE}(P^\star)\big),$$

where

$$\mathrm{PNDE}(P^\star) = \mathbb{E}_{P^\star}\{Y(1, M(0)) - Y(0, M(0))\}, \qquad \mathrm{TNIE}(P^\star) = \mathbb{E}_{P^\star}\{Y(1, M(1)) - Y(1, M(0))\}.$$

The identified set is

$$\Theta_I(P) = \{\theta(P^\star) : P^\star \text{ is finite, satisfies Assumptions 1–6, and induces } P\}.$$

This definition distinguishes the causal target from any bridge, regression, or LP functional computed from the observed law.

For each positive-probability (a, m, z, x), define

$$K_{a,m,x}(z, w) = P(W = w \mid A = a, M = m, Z = z, X = x), \qquad q_{a,m,x}(z) = \mathbb{E}(Y \mid A = a, M = m, Z = z$$

The bridge equation is

$$K_{a,m,x} h_{a,m,x} = q_{a,m,x},$$

where $h_{a,m,x} \in [0, 1]^2$ is indexed by $w \in \{0, 1\}$. Also define

$$\pi_{r,m,x} = P(M = m \mid A = r, X = x), \qquad \omega_{a,m,x}(w) = P(W = w \mid A = a, M = m, X = x).$$

Because $\omega_{a,m,x}$ averages over Z, it satisfies

$$\omega_{a,m,x}(w) = \sum_{z=0}^{1} P(Z = z \mid A = a, M = m, X = x) K_{a,m,x}(z, w).$$

This row-span identity is the key algebraic fact behind the refutation below.

When $K_{a,m,x}$ has rank one, choose a reference bridge solution $h^0_{a,m,x}$ and a nonzero null vector $\nu_{a,m,x}$ with $K_{a,m,x}\nu_{a,m,x} = 0$. The bounded bridge interval is

$$T_{a,m,x}(P) = \{t \in \mathbb{R} : 0 \le h^0_{a,m,x}(w) + t\nu_{a,m,x}(w) \le 1 \text{ for } w = 0, 1\}.$$

The set of rank-one cells is denoted $\mathcal{N}$.

Locked parameters.

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qid = pid_mediation_proxy
specialization = v1
anchor_cluster = PartialID / mediation bounds with negative-control proxies
observable distribution P = law(A,M,Y,Z,W,X) with A,M,Y,Z,W binary and X finite
parameter theta = (PNDE,TNIE), where
  PNDE = E[Y(1,M(0))-Y(0,M(0))]
  TNIE = E[Y(1,M(1))-Y(1,M(0))]
identifying restrictions =
  consistency and bounded outcomes; randomized single binary treatment A;
  finite latent post-treatment mediator-outcome confounder U;
  negative-control proxy exclusions for binary treatment-inducing proxy Z
    and binary outcome-inducing proxy W;
  latent mediator-outcome exchangeability given (A,U,X);
  positivity for all observed (A,M,Z,W,X) cells used in the bridge equations;
  existence of finite outcome-confounding bridge arrays h_{a,m,x}(w);
  proxy-rank regime rank(K_{a,m,x}) in {1,2}, where
    K_{a,m,x}(z,w)=P(W=w | A=a,M=m,Z=z,X=x).
bounding functional =
  the sharp joint projection of the finite latent/proxy-response feasible
  set onto (PNDE,TNIE); in the rank-one proxy-null regime the proposed
  closed form is the two-dimensional affine zonotope generated by the
  bridge-null intervals and their PNDE/TNIE loading vectors, with
  observable proxy-separation inequalities certifying strict improvement
  over no-proxy mediation bounds.
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4 Assumptions

Assumption 1 (Finite support, bounded outcomes, and positivity)**.** The variables have the domains stated in Section 3, X has finite support, and $Y \in \{0,1\}$. For every x with $P(X = x) > 0$, $0 < P(A = a \mid X = x) < 1$ for $a = 0, 1$. Every conditional probability and conditional mean used in $K_{a,m,x}$, $q_{a,m,x}$, $\pi_{r,m,x}$, and $\omega_{a,m,x}$ is evaluated only on cells with positive denominator; the locked positivity condition requires positive observed (A, M, Z, W, X) cells used in the bridge equations.

Assumption 2 (Consistency and randomized treatment)**.** The factual mediator and outcome satisfy $M = M(A)$ and $Y = Y(A, M(A))$. The single binary treatment A is randomized conditional on X: A is independent of all mediator, outcome, proxy, and latent response types given X.

Assumption 3 (Finite latent mediator-outcome exchangeability). There exists a finite latent variable U such that, conditional on (A, U, X), the mediator assignment carries no residual information about the nested outcome response beyond U, and, conditional on (A, M, U, X), the outcome response is independent of the observed treatment-inducing proxy Z. This is the maintained latent mediator-outcome exchangeability condition in the Stage -1 proposal.

Assumption 4 (Negative-control proxy exclusions). The proxy Z has no direct effect on Y after conditioning on (A, M, U, X). The proxy W is not affected by interventions on M after conditioning on (A, U, X). Conditional on (A, M, U, X), Z and W are independent except through their common dependence on U.

Assumption 5 (Observable bridge feasibility and boundedness). For every positive-probability (a, m, x), the observable bridge equation

$$K_{a,m,x}h_{a,m,x} = q_{a,m,x}$$

has at least one solution $h_{a,m,x} \in [0, 1]^2$. In rank-one cells the reference solution $h^0_{a,m,x}$, null vector $\nu_{a,m,x}$, and interval $T_{a,m,x}(P)$ are those defined in Section 3.

Assumption 6 (Proxy-rank regime). For each (a, m, x) with positive denominator, the 2×2 matrix $K_{a,m,x}$ has rank one or rank two. Rank-two cells are complete observed bridge cells, and rank-one cells are proxy-null cells.

Assumption 7 (No-proxy and proxy endpoint faces). The finite proxy-response polytope $\mathcal{L}^{\text{proxy}}(P)$, the no-proxy polytope $\mathcal{L}^0(P)$, the forgetting map $\rho : \mathcal{L}^{\text{proxy}}(P) \to \mathcal{L}^0(P)$, and the no-proxy endpoint faces $F^0_{E,s}(P)$ for $E \in \{D, I\}$ and $s \in \{\min, \max\}$ are the finite objects defined in Section 5. The observable proxy endpoint hull $C_{E,s}(P)$, separation margin $\Delta_{E,s}(P)$, endpoint reduction $R_{E,s}(P)$, and finite Lipschitz constant $L_{E,s}(P)$ are also those defined in Section 5.

Assumption 8 (Sequential-ignorability corner). There is no latent mediator-outcome confounding given (A, M, X), and the usual sequential-ignorability mediation formula is valid for PNDE and TNIE. This is used only as a sanity-check specialization, not as a maintained assumption for the main partial-identification results.

5 Regression estimator

This section title is fixed by the skeleton. In this partial-identification anchor there is no regression estimator. The estimand-defining functional is the bounding functional.

Definition 1 (Observable bridge plug-in map)**.** For a bridge array $h = \{h_{a,m,x}\}$, define

$$\Psi_a(r;h) = \sum_{x\in\mathcal{X}} P(X=x) \sum_{m=0}^{1} \pi_{r,m,x} \sum_{w=0}^{1} \omega_{a,m,x}(w) h_{a,m,x}(w).$$

The bridge plug-in natural-effect coordinates are

$$D_{\mathrm{br}}(h) = \Psi_1(0;h) - \Psi_0(0;h), \qquad I_{\mathrm{br}}(h) = \Psi_1(1;h) - \Psi_1(0;h).$$

The Stage -1 conjectured bridge-zonotope is the image of the observable bridge solution set under $h \mapsto (D_{\mathrm{br}}(h), I_{\mathrm{br}}(h))$.

For a rank-one cell $j = (a,m,x)$, write

$$\eta_j(P) = \sum_{w=0}^{1} \omega_{a,m,x}(w)\nu_{a,m,x}(w).$$

The proposed PNDE and TNIE loadings are

$$\lambda^D_{a,m,x} = P(X=x)\pi_{0,m,x}\eta_{a,m,x}\{\mathbf{1}(a=1) - \mathbf{1}(a=0)\},$$

and

$$\lambda^I_{a,m,x} = P(X=x)\mathbf{1}(a=1)(\pi_{1,m,x} - \pi_{0,m,x})\eta_{1,m,x}.$$

The conjectured set is

$$Z_{\mathrm{br}}(P) = \left\{ (D_{\mathrm{br}}(h^0), I_{\mathrm{br}}(h^0)) + \sum_{j\in\mathcal{N}} t_j(\lambda^D_j, \lambda^I_j) : t_j \in T_j(P) \right\}.$$

Theorem 1 proves that, with the stated K and ω, these loadings are zero in every rank-one cell.

Definition 2 (Finite proxy-response LP bounding functional)**.** Let $\mathcal{R}_x$ be the finite set of deterministic response types at baseline covariate value x. A response type records all potential mediator, outcome, and proxy coordinates needed to evaluate the observed data under $A = 0, 1$ and to evaluate

$Y(1,M(0))$, $Y(0,M(0))$, and $Y(1,M(1))$, with the exclusion restrictions in Assumption 4 imposed as coordinate equalities. Let $\mu_x(r) \geq 0$ be the mass assigned to $r \in \mathcal{R}_x$. The proxy-response feasible polytope is

$$\mathcal{L}^{\text{proxy}}(P) = \{\mu : \mu_x(r) \geq 0,\ A_{\text{obs}}\mu = b^{\text{proxy}}(P),\ A_{\text{res}}\mu = 0\},$$

where $A_{\text{obs}}\mu = b^{\text{proxy}}(P)$ matches the observed cell law of (A, M, Y, Z, W, X), and $A_{\text{res}}\mu = 0$ collects the deterministic equalities implied by the maintained restrictions. Define the linear functionals

$$D(\mu) = \sum_{x,r} \mu_x(r)\{Y_r(1, M_r(0)) - Y_r(0, M_r(0))\},$$

$$I(\mu) = \sum_{x,r} \mu_x(r)\{Y_r(1, M_r(1)) - Y_r(1, M_r(0))\}.$$

The sharp LP joint set and coordinate bounds are

$$\Theta_{\text{LP}}(P) = \{(D(\mu), I(\mu)) : \mu \in \mathcal{L}^{\text{proxy}}(P)\},$$

$$\theta_L^E(P) = \min_{\mu \in \mathcal{L}^{\text{proxy}}(P)} E(\mu), \qquad \theta_U^E(P) = \max_{\mu \in \mathcal{L}^{\text{proxy}}(P)} E(\mu), \quad E \in \{D, I\}.$$

Definition 3 (No-proxy comparison and endpoint separation)**.** Let $\mathcal{L}^0(P)$ be the no-proxy finite response-type polytope obtained after dropping Z, W and matching only the observed law of (A, M, Y, X). Let $\rho : \mathcal{L}^{\text{proxy}}(P) \to \mathcal{L}^0(P)$ forget the proxy response coordinates. For $E \in \{D, I\}$ and $s \in \{\min, \max\}$, let $F^0_{E,s}(P)$ be the exposed face of $\mathcal{L}^0(P)$ attaining the corresponding no-proxy endpoint. Let $C_{E,s}(P)$ be the convex hull of observable proxy cell-probability vectors generated by no-proxy endpoint completions in $F^0_{E,s}(P)$, and define

$$\Delta_{E,s}(P) = \sup_{\alpha : \|\alpha\|_1 \leq 1} \left\{ \alpha^\top b^{\text{proxy}}(P) - \sup_{c \in C_{E,s}(P)} \alpha^\top c \right\}_+ .$$

The endpoint reduction $R_{E,s}(P)$ is the no-proxy endpoint minus the proxy-LP endpoint in the direction s, so it is nonnegative when proxies tighten the bound.

6 Target causal estimand

The intended target is not a bridge solution and not an LP optimum. It is the causal vector

$$\theta(P^\star) = \big(\mathbb{E}_{P^\star}\{Y(1, M(0)) - Y(0, M(0))\}, \mathbb{E}_{P^\star}\{Y(1, M(1)) - Y(1, M(0))\}\big).$$

The first coordinate is the pure natural direct effect: compare treatment levels while holding the mediator to the value it would have taken under $A = 0$. The second coordinate is the total natural indirect effect: compare the mediator under $A = 1$ to the mediator under $A = 0$, while holding the outcome treatment level at $A = 1$. Partial identification asks which values of this vector can be generated by finite latent causal laws $P^\star$ that induce the same observed law P and satisfy the maintained restrictions.

7 Main theorems

Proposition 1 (Rank-one observed bridge-null loadings vanish)**.** *Under Assumptions 1, 5, and 6, if $v \in \mathbb{R}^2$ satisfies $K_{a,m,x}v = 0$, then*

$$\sum_{w=0}^{1} \omega_{a,m,x}(w)v(w) = 0.$$

Consequently every rank-one null loading $(\lambda^D_{a,m,x}, \lambda^I_{a,m,x})$ defined in Section 5 is zero, and $Z_{\mathrm{br}}(P)$ is a singleton.

This proposition is a deterministic row-span fact, not a new identification theorem. It is important because it already contradicts the nontrivial proxy-null phase diagram proposed in Stage -1: with the locked K and ω, rank-one bridge nonuniqueness cannot by itself create nonzero PNDE or TNIE bridge loadings.

Theorem 1 (Conjecture 1 is refuted as a causal sharp set)**.** *Under the locked assumptions as stated in Sections 3 and 4, the Stage -1 Conjecture 1 is false if read as a sharp causal identified set for $\Theta_I(P)$. There exists an observed law P satisfying Assumptions 1–6 such that every $K_{a,m,x}$ is rank one, $Z_{\mathrm{br}}(P) = \{(0, 0)\}$, but*

$$(-1/4,\, 1/4) \in \Theta_I(P) \quad \textit{and} \quad (1/4,\, -1/4) \in \Theta_I(P).$$

Thus the bridge-zonotope is not an outer bound for the causal target, and the advertised missing-corner geometry is not the sharp joint PNDE/TNIE set.

The novelty here is the negative conclusion. The nearest correct bridge-only statement is that the bridge solution set maps to a degenerate singleton in the rank-one binary conditional-proxy case. That bridge-image singleton is not a causal bound unless an additional latent bridge or completeness condition is added as a locked restriction. This differs from proximal point-identification results such as [Dukes, Shpitser, and Tchetgen Tchetgen(2023)], where bridge conditions are strong enough to connect the observable bridge equation to the latent target.

Theorem 2 (Correct sharp bound and corrected proxy-separation verdict)**.** *Under Assumptions 1–4, the finite proxy-response LP is sharp:*

$$\Theta_I(P) = \Theta_{\mathrm{LP}}(P).$$

Hence

$$\Theta_I(P) \subseteq [\theta_L^D(P), \theta_U^D(P)] \times [\theta_L^I(P), \theta_U^I(P)],$$

with equality to the displayed rectangle only when the two coordinate extrema are jointly attainable. For each no-proxy endpoint (E, s), strict tightening of that endpoint by the proxy LP occurs if and only if

$$\rho(\mathcal{L}^{\mathrm{proxy}}(P)) \cap F_{E,s}^0(P) = \varnothing.$$

Equivalently, in the finite endpoint-hull notation of Definition 3, strict tightening occurs if and only if $\Delta_{E,s}(P) > 0$. The endpoint reduction $R_{E,s}(P)$ is exactly the difference between the no-proxy endpoint and the proxy-LP endpoint; when $L_{E,s}(P) > 0$, every separating contrast gives the computable certificate $R_{E,s}(P) \geq \Delta_{E,s}(P)/L_{E,s}(P)$.

This theorem partially confirms the spirit of Stage -1 Conjecture 2 but refutes it as stated. The observable separation criterion is correct for finite LP endpoint faces, by the same convex-separation logic used in Manski and Balke–Pearl style bounds. It is not a certificate for the coordinate projections of the refuted bridge-zonotope formula.

8 Interpretation

The first proposition says that the rank-one bridge-null mechanism in the proposal is algebraically inert. Since $P(W \mid A = a, M = m, X = x)$ is a mixture of the rows of $P(W \mid A = a, M = m, Z = z, X = x)$, any vector invisible to all rows is also invisible to the target weight used by the proposed bridge plug-in formula. Thus a binary rank-one K cannot produce the advertised nonzero loading vectors.

The refutation theorem says something stronger. Even the degenerate bridge-image singleton need not contain the causal natural effects. The observable bridge equation balances $E(Y \mid A, M, Z, X)$ against a function of W, but without a completeness or latent-bridge condition it does not pin down the cross-world outcome values $Y(1, M(0))$ and $Y(1, M(1))$. The contamination source is not a regression coefficient or a nonzero bridge-null loading; it is latent response-type variation that is invisible in the factual observed cells but visible in the natural-effect counterfactuals.

The corrected LP theorem is the practical bound. It keeps all finite latent response types consistent with the observed distribution and the proxy restrictions, then projects that feasible set onto the two causal coordinates. The joint set need not be a rectangle, so separately sharp PNDE and TNIE interval endpoints should not be paired unless the LP confirms joint attainability. Proxy variables tighten a no-proxy endpoint exactly when the observed proxy distribution rules out every no-proxy response-type completion that would attain that endpoint.

9 Step-by-step proof

Proof of Proposition 1. Fix (a, m, x) and write

$$\gamma_z = P(Z = z \mid A = a, M = m, X = x).$$

By Assumption 1, these conditional probabilities are defined on the cells under consideration. The law of total probability gives, for each w,

$$\omega_{a,m,x}(w) = P(W = w \mid A = a, M = m, X = x) = \sum_{z=0}^{1} \gamma_z K_{a,m,x}(z, w).$$

Let v satisfy $K_{a,m,x} v = 0$. Then

$$\sum_w \omega_{a,m,x}(w) v(w) = \sum_w \sum_z \gamma_z K_{a,m,x}(z, w) v(w) = \sum_z \gamma_z \sum_w K_{a,m,x}(z, w) v(w) = 0.$$

In a rank-one cell, all bridge solutions have the form $h^0_{a,m,x} + t\nu_{a,m,x}$ by Assumptions 5 and 6. Taking $v = \nu_{a,m,x}$ gives $\eta_{a,m,x} = 0$, hence the PNDE and TNIE loadings in Section 5 are both zero. Therefore all rank-one terms in $Z_{\mathrm{br}}(P)$ vanish and the bridge image is the singleton generated by any reference solution.

Proof of Theorem 1. It is enough to construct two finite causal laws that induce the same observed law P, satisfy the maintained restrictions, and have different natural effects while the bridge-zonotope is $\{(0, 0)\}$. Take X to be degenerate, $A \sim \text{Bernoulli}(1/2)$, and four latent response types

$r \in \{n, c, d, e\}$, each with probability $1/4$. Their mediator potential values are

r	n	c	d	e
$M(0)$	0	0	1	1
$M(1)$	0	1	0	1.

In both causal laws set the factual outcome coordinates

$$Y_{00}(n) = 0, \quad Y_{00}(c) = 1, \quad Y_{01}(d) = 0, \quad Y_{01}(e) = 1,$$

$$Y_{10}(n) = 0, \quad Y_{10}(d) = 1, \quad Y_{11}(c) = 0, \quad Y_{11}(e) = 1.$$

All unlisted potential outcomes are set to zero except for the two coordinates $Y_{10}(c)$ and $Y_{11}(d)$. In law $P^\star_-$, set

$$Y_{10}(c) = 0, \qquad Y_{11}(d) = 0.$$

In law $P^\star_+$, set

$$Y_{10}(c) = 1, \qquad Y_{11}(d) = 1.$$

Let Z and W be independent fair coins, independent of all response types. Then the consistency and randomized-treatment requirements in Assumption 2 hold. The finite latent $U = r$ makes the mediator and outcome response deterministic conditional on U, so Assumption 3 holds. The proxies have no direct effects and are conditionally independent given U, so Assumption 4 holds.

Both laws induce the same observed distribution. In each of the four factual (A, M) cells, the two response types in that cell have observed outcome values 0 and 1, so

$$P(Y = 1 \mid A = a, M = m, Z = z, W = w) = 1/2$$

for all a, m, z, w. Since Z, W are fair independent coins,

$$K_{a,m}(z, w) = 1/2 \quad \text{for all } a, m, z, w,$$

so every $K_{a,m}$ has rank one. The bridge equation is feasible because $h_{a,m}(0) = h_{a,m}(1) = 1/2$ solves it. Proposition 1 implies $Z_{\mathrm{br}}(P) = \{(0, 0)\}$.

The two causal laws have different natural effects. For $P^\star_-$,

$$\mathrm{PNDE} = \frac{1}{4}\{0 + (0 - 1) + (0 - 0) + (1 - 1)\} = -\frac{1}{4},$$

and

$$\mathrm{TNIE} = \frac{1}{4}\{0 + (0 - 0) + (1 - 0) + 0\} = \frac{1}{4}.$$

For $P^\star_+$,

$$\mathrm{PNDE} = \frac{1}{4}\{0 + (1-1) + (1-0) + (1-1)\} = \frac{1}{4},$$

and

$$\mathrm{TNIE} = \frac{1}{4}\{0 + (0-1) + (1-1) + 0\} = -\frac{1}{4}.$$

Thus both $(-1/4, 1/4)$ and $(1/4, -1/4)$ belong to $\Theta_I(P)$, while $Z_{\mathrm{br}}(P) = \{(0,0)\}$. The conjectured bridge-zonotope is therefore not an outer bound and cannot be sharp.

Proof of Theorem 2. The proof is the finite extreme-point argument used in Manski and Balke–Pearl linear-programming bounds. Because all variables and X have finite support by Assumption 1, every finite causal law satisfying Assumptions 2–4 can be written as a mixture of deterministic response types. The observed distribution imposes linear equality constraints on the mixture weights, and the proxy exclusions impose coordinate equalities. Therefore every compatible $P^\star$ maps to some $\mu \in \mathcal{L}^{\mathrm{proxy}}(P)$, and its natural-effect vector is $(D(\mu), I(\mu))$. This proves

$$\Theta_I(P) \subseteq \Theta_{\mathrm{LP}}(P).$$

Conversely, given any $\mu \in \mathcal{L}^{\mathrm{proxy}}(P)$, construct a finite causal law by first drawing X and a deterministic response type with probability $\mu_x(r)$, then drawing the randomized treatment A according to its observed conditional distribution and revealing the corresponding factual variables by consistency. The linear constraints guarantee that this constructed law induces P and satisfies the proxy restrictions. Its natural-effect vector is exactly $(D(\mu), I(\mu))$, proving

$$\Theta_{\mathrm{LP}}(P) \subseteq \Theta_I(P).$$

The equality $\Theta_I(P) = \Theta_{\mathrm{LP}}(P)$ follows.

The coordinate bounds are minima and maxima of linear functionals over the compact polytope $\mathcal{L}^{\mathrm{proxy}}(P)$, so they are attained at faces of the polytope. The displayed rectangle contains the joint set because each coordinate of any feasible point lies between its coordinate minimum and maximum. Equality to the rectangle requires every coordinate pair in the rectangle to be generated by some common feasible μ, which is precisely joint attainability.

For the endpoint comparison, a no-proxy endpoint survives the proxy restrictions if and only if some proxy-feasible law forgets to a point on the no-proxy endpoint face:

$$\rho(\mathcal{L}^{\mathrm{proxy}}(P)) \cap F^0_{E,s}(P) \neq \varnothing.$$

If this intersection is nonempty, the proxy endpoint equals the no-proxy endpoint in direction s. If it is empty, compactness and convexity imply a strict endpoint reduction. The endpoint-completing proxy cell vectors form the closed convex hull $C_{E,s}(P)$, so the intersection is empty exactly when the observed proxy vector $b^{\text{proxy}}(P)$ lies outside this hull. By the finite-dimensional separating hyperplane theorem, this is equivalent to the existence of a contrast α with

$$\alpha^{\mathsf{T}} b^{\text{proxy}}(P) > \sup_{c \in C_{E,s}(P)} \alpha^{\mathsf{T}} c,$$

which is equivalent to $\Delta_{E,s}(P) > 0$ after normalizing $\|\alpha\|_1 \leq 1$. The exact value $R_{E,s}(P)$ is the difference between two LP optima by definition. If $L_{E,s}(P) > 0$, the stated Lipschitz property converts the separating distance into the lower certificate $R_{E,s}(P) \geq \Delta_{E,s}(P)/L_{E,s}(P)$. If $L_{E,s}(P) = 0$, the separation criterion remains valid but this ratio certificate is not a meaningful finite number.

10 Sanity checks

No-proxy fallback. If Z, W and Assumption 4 are dropped, the bounding functional is the projection of $\mathcal{L}^0(P)$ onto PNDE and TNIE. This is the natural Manski-style finite mediation bound: all counterfactual response types that match the observed law of (A, M, Y, X) are allowed, and the endpoints are LP optima.

Zero-width point-identification corner. If an additional valid point-identification condition makes $\Theta_I(P)$ a singleton, then the LP coordinate bounds collapse:

$$\theta_L^D(P) = \theta_U^D(P), \qquad \theta_L^I(P) = \theta_U^I(P).$$

Sequential ignorability in Assumption 8 is one such sanity-check condition, yielding the usual mediation formula of [Imai, Keele, and Yamamoto(2010)]. A proximal latent bridge plus sufficient completeness is another, matching the point-identification logic of [Dukes, Shpitser, and Tchetgen Tchetgen(2023)]. The locked assumptions in this note do not include that extra latent bridge/completeness condition.

Rank-one proxy corner. The proposed rank-one bridge formula should have reduced to a nontrivial zonotope if the Stage -1 conjecture were right. Proposition 1 shows the opposite for the locked K: every null loading is zero. The counterexample in Theorem 1 then shows that the causal set can still be non-singleton, so the loss of identification is not captured by the bridge-null loading vectors.

MTR specialization. Monotone treatment response is not part of the locked parameter block. If it were added as a new restriction, it would enter the finite LP as additional linear inequalities such as $Y_r(1,m) \geq Y_r(0,m)$. That would weakly tighten $\Theta_{\mathrm{LP}}(P)$, but it is outside the theorem proved here.

Balke–Pearl finite-LP special case. When the target is changed from natural mediation effects to a binary treatment effect with a binary IV and finite response types, the construction reduces to the Balke–Pearl LP template [Balke and Pearl(1997)]. The present theorem uses the same finite sharpness logic, but its response types include mediator and cross-world natural-effect coordinates.

11 Counterexample or minimality check

The counterexample in the proof of Theorem 1 is the minimality check for the main negative result. It has one X stratum, binary variables, four mediator principal strata, independent fair negative-control proxies, and positive observed (A, M, Z, W) cells. The observed bridge matrices are all

$$K_{a,m} = \begin{pmatrix} 1/2 & 1/2 \\ 1/2 & 1/2 \end{pmatrix},$$

so they are rank one and bridge feasible. The bridge solution set is nonunique, but the proposed target weights average over the same row space, hence $Z_{\mathrm{br}}(P) = \{(0,0)\}$.

The two compatible causal laws differ only in the unobserved cross-world outcome coordinates $Y(1,0)$ for mediator compliers and $Y(1,1)$ for mediator defiers. These coordinates do not change the factual observed law, but they do change $Y(1, M(0))$ and $Y(1, M(1))$. Therefore the observable bridge equation cannot be the only condition connecting proxies to the natural-effect target.

The nearest correct statement is that the affine bridge-null construction describes only the image of observable bridge solutions, not the sharp causal identified set. For the locked binary conditional-proxy matrix, even that bridge image is a degenerate singleton in rank-one cells. To turn a bridge formula into a causal sharp bound, one must either add a latent bridge/completeness restriction as a locked assumption or use the finite proxy-response LP projection $\Theta_{\mathrm{LP}}(P)$.

12 Formalization checklist

Probabilistic objects.

- Observable finite law:

$$P : \{0,1\}^5 \times \mathcal{X} \to [0,1], \qquad \sum_{a,m,y,z,w,x} p(a,m,y,z,w,x) = 1.$$

  Role: supplies all observable conditionals and LP right-hand sides.

- Latent causal law $P^\star$: finite distribution over X, U and deterministic response types for $M(a)$, $Y(a,m)$, Z, and W, subject to consistency, randomization, and proxy-exclusion restrictions. Role: defines $\Theta_I(P)$.

- Natural-effect functionals:

$$D(P^\star) = \mathbb{E}\{Y(1,M(0)) - Y(0,M(0))\}, \quad I(P^\star) = \mathbb{E}\{Y(1,M(1)) - Y(1,M(0))\}.$$

  Role: target coordinates.

- Observable bridge matrices and means:

$$K_{a,m,x}(z,w) = P(W = w \mid A = a, M = m, Z = z, X = x), \quad q_{a,m,x}(z) = \mathbb{E}(Y \mid A = a, M = m, Z = z$$

  Role: define the refuted bridge-null formula.

Deterministic identities and convex objects.

- Row-span identity:

$$\omega_{a,m,x} = \sum_z P(Z = z \mid A = a, M = m, X = x) K_{a,m,x}(z, \cdot).$$

  Dependency: support and the law of total probability. Role: proves null loadings vanish.

- Null-loading identity: if $K_{a,m,x}v = 0$, then $\omega_{a,m,x}v = 0$. Dependency: row-span identity. Role: collapses the rank-one bridge-zonotope.

- Finite response-type polytope:

$$\mathcal{L}^{\text{proxy}}(P) = \{\mu \geq 0 : A_{\text{obs}}\mu = b^{\text{proxy}}(P), A_{\text{res}}\mu = 0\}.$$

  Role: sharp feasible set for all compatible finite causal laws.

- Linear target maps $D(\mu)$ and $I(\mu)$. Dependency: response-type coordinates for $M(0), M(1), Y(0,m), Y(1,m)$. Role: projection defining $\Theta_{\text{LP}}(P)$.

- Sharpness lemma: finite compatible causal laws and feasible LP mixtures are equivalent by deterministic response-type expansion. Role: proves $\Theta_I(P) = \Theta_{\text{LP}}(P)$.

- Convex separation lemma: for compact convex endpoint hull $C_{E,s}(P)$, $b^{\text{proxy}}(P) \notin C_{E,s}(P)$ iff a normalized contrast α yields $\Delta_{E,s}(P) > 0$. Role: corrected proxy tightening certificate.

- Counterexample response table: four mediator principal strata with masses 1/4, independent fair proxies, and two alternative assignments of $(Y_{10}(c), Y_{11}(d))$. Role: refutes Conjecture 1 as a causal bound.

13 Conclusion and references

The theorem-level contribution is a correction of the Stage -1 proposal. The rank-one bridge-null zonotope is not a sharp causal PNDE/TNIE identified set under the locked assumptions; in fact, the proposed rank-one loadings are zero because the target W-weights lie in the row span of the observed bridge matrix. The main source of nonidentification is not a nonzero bridge-null coefficient but unobserved response-type variation in cross-world natural-effect coordinates. The sharp replacement is the finite proxy-response LP projection, with strict no-proxy endpoint tightening characterized by finite convex separation of endpoint faces.

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